

OCR Further Mathematics
Mechanics - Center of Mass
Question Paper
AS and A Level
Further Mathematics A - H235, H245



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The latest topic questions for the year 2024 / 2025+ exams.
To be used by all students preparing for OCR AS and A Level Further Mathematics.
Students of other boards may also find this useful.

- 1(a) A rectangular lamina of mass M has vertices at the origin $O(0, 0)$, $A(24a, 0)$, $B(24a, 6a)$ and $C(0, 6a)$, where a is a positive constant. A small object P of mass m is attached to the lamina at the point (x, y) . The centre of mass of the system consisting of the lamina and P is at the point $(\bar{x}, \bar{y})$. P is modelled as a particle and the lamina is modelled as being uniform.

Show that $\bar{x} = \frac{12Ma + mx}{M + m}$. [1]

- (b) Find a corresponding expression for $\bar{y}$. [1]

- (c) The lamina, with P no longer attached, is placed on a horizontal rectangular table, with its sides parallel to the edges of the table, and partly overhanging the edges of the table, as shown in the diagram. The corner of the table is at the point $(6a, 2a)$.



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© When P is placed on the lamina at O , the lamina topples over one of the edges of the table.

Show that $m > \frac{1}{2}M$. [4]

- (d) The lamina is now put back on the table in the same position as before. P is placed at the point $(12a, 6a)$ on the smooth upper surface of the lamina, and is projected towards O . At a subsequent instant during the motion, P is at the point $(12ak, 6ak)$ where $0 < k < 1$.

Assuming that the lamina has not yet toppled, find, in terms of M and m , the value of k for which the centre of mass of the system lies on the table edge parallel to OC . [3]

(e)

For the case $m = \frac{3}{2}M$, determine which table edge the lamina topples over.

[4]

- 2(a) Fig. 7.1 shows a uniform lamina in the shape of a sector of a circle of radius r and angle 2θ where θ is in radians. The sector consists of a triangle OAB and a segment bounded by the chord AB .

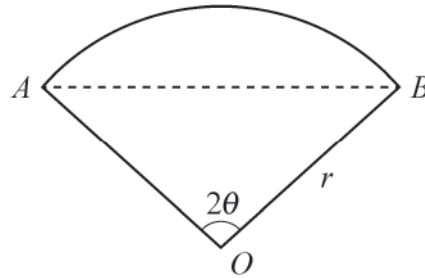


Fig. 7.1

Explain why the centre of mass of the segment lies on the radius through the midpoint of AB .

[1]

(b)

Show that the distance of the centre of mass of the segment from O is $\frac{2r \sin^3 \theta}{3(\theta - \sin \theta \cos \theta)}$.

[4]

- (c) A uniform circular lamina of radius 5 units is placed with its centre at the origin, O , of an x - y coordinate system. A component for a machine is made by removing and discarding a segment from the lamina. The radius of the circle from which the segment is formed is 3 units and the centre of this circle is O . The centre of the straight edge of the segment has coordinates $(0, 2)$ and this edge is perpendicular to the y -axis (see Fig. 7.2).

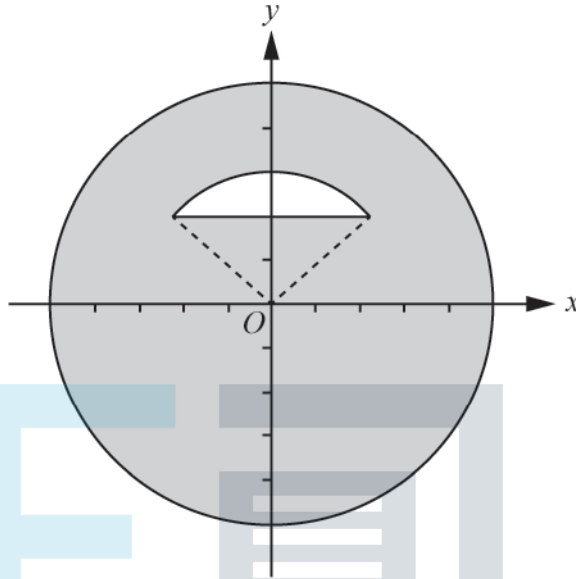


Fig. 7.2

Find the y -coordinate of the centre of mass of the component, giving your answer correct to 3 significant figures.

[5]

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- (d) C is the point on the component with coordinates $(0, 5)$. The component is now placed horizontally and supported only at O . A particle of mass m kg is placed on the component at C and the component and particle are in equilibrium.

Find the mass of the component in terms of m .

[2]

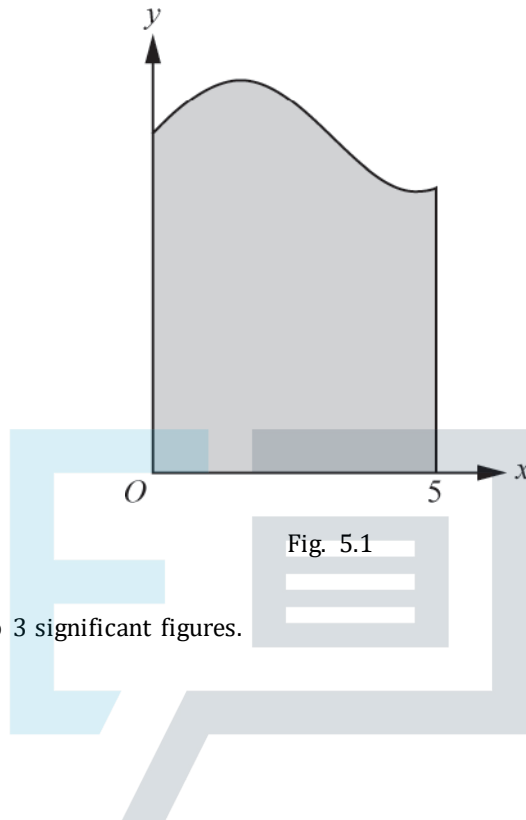


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- 3(a) The cover of a children's book is modelled as being a uniform lamina L . L occupies the region bounded by the x -axis, the curve $y = 6 + \sin x$ and the lines $x = 0$ and $x = 5$ (see Fig. 5.1). The centre of mass of L is at the point $(\bar{x}, \bar{y})$.



Show that $\bar{x} = 2.36$, correct to 3 significant figures.

[2]

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- (b) Find $\bar{y}$, giving your answer correct to 3 significant figures.

[2]

- (c) The cover of the book weighs 6 N. A is the point on the cover with coordinates $(3, \bar{y})$ and B is the point on the cover with coordinates $(5, \bar{y})$. A small badge of weight 2 N is attached to the cover at A .

The side of L along the y -axis is attached to the rest of the book and the book is placed on a rough horizontal plane. The attachment of the cover to the book is modelled as a hinge.

The cover is held in equilibrium at an angle of $\frac{1}{3}\pi$ radians to the horizontal by a force of magnitude P N acting at B perpendicular to the cover (see Fig. 5.2).

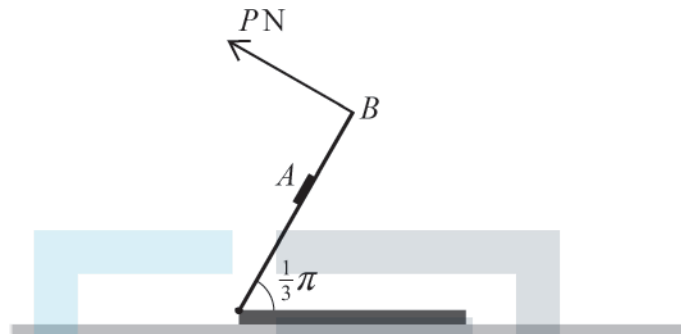


Fig. 5.2

State two additional modelling assumptions, one about the attachment of the cover and one about the badge, which are necessary to allow the value of P to be determined.

[2]

- (d) Using the modelling assumptions, determine the value of P giving your answer correct to 3 significant figures.

[3]

4(a)

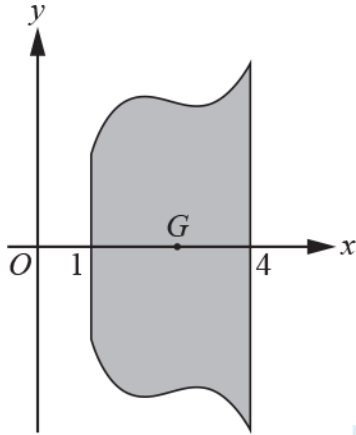


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The region bounded by the x -axis, the curve $y = \sqrt{2x^3 - 15x^2 + 36x - 20}$ and the lines $x = 1$ and $x = 4$ is rotated through 2π radians about the x -axis to form a uniform solid of revolution R . The centre of mass of R is the point G (see diagram).



Explain why the y -coordinate of G is 0.

[1]

(b) Find the x -coordinate of G .

[4]



(c) P is a point on the edge of the curved surface of R where $x = 4$. R is freely suspended from P and hangs in equilibrium.

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Find the angle between the axis of symmetry of R and the vertical.

[3]

- 5 Particles A , B and C of masses 2 kg, 3 kg and 5 kg respectively are joined by light rigid rods to form a triangular frame. The frame is placed at rest on a horizontal plane with A at the point $(0, 0)$, B at the point $(0.6, 0)$ and C at the point $(0.4, 0.2)$, where distances in the coordinate system are measured in metres (see Fig. 1).

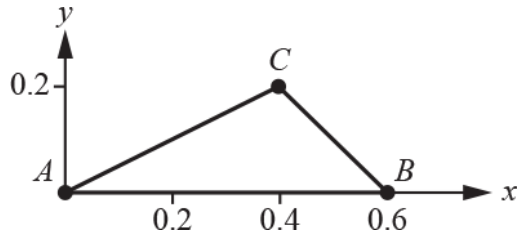


Fig. 1

G , which is the centre of mass of the frame, is at the point $(\bar{x}, \bar{y})$.

- (a)
- Show that $\bar{x} = 0.38$,
 - Find $\bar{y}$.

[3]

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- (b) Explain why it would be impossible for the frame to be in equilibrium in a horizontal plane supported at only one point.

[1]

A rough plane, Π , is inclined at an angle i to the horizontal where $\sin \theta = \frac{3}{5}$. The frame is placed on Π with AB vertical and B in contact with Π . C is in the same vertical plane as AB and a line of greatest slope of Π . C is on the down-slope side of AB .

The frame is kept in equilibrium by a horizontal light elastic string whose natural length is l m and whose modulus of elasticity is g N. The string is attached to A at one end and to a fixed point on Π at the other end (see Fig. 2).

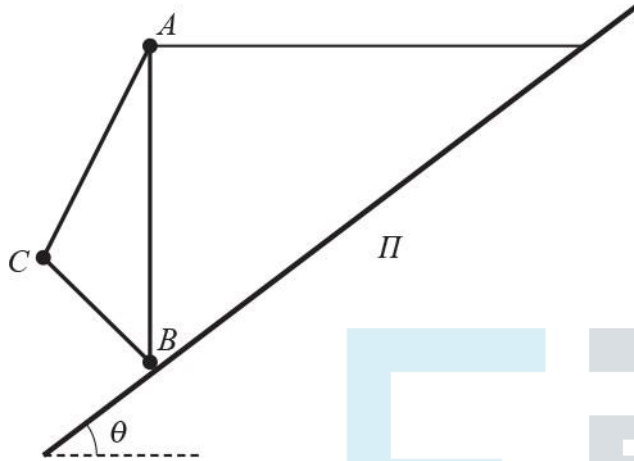


Fig. 2

The coefficient of friction between B and Π is μ .

(c) Show that $l = 0.3$.

[3]

(d) Show that $\mu \geq \frac{14}{27}$.

[6]



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- 6 A uniform solid hemisphere has radius 0.4 m. A uniform solid cone, made of the same material, has base radius 0.4 m and height 1.2 m. A solid, S , is formed by joining the hemisphere and the cone so that their circular faces coincide. O is the centre of the joint circular face and V is the vertex of the cone. G is the centre of mass of S .

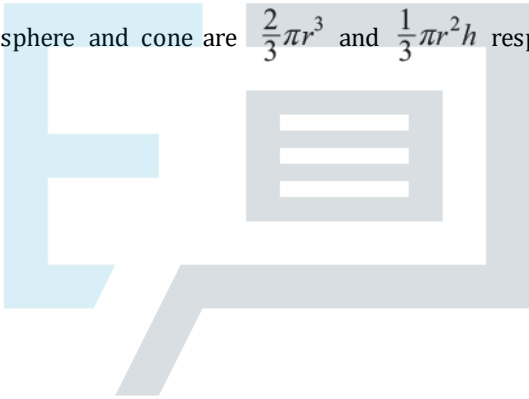
(a) Explain briefly why G lies on the line through O and V .

[1]

(b) Show that the distance of G from O is 0.12 m.

(The volumes of a hemisphere and cone are $\frac{2}{3}\pi r^3$ and $\frac{1}{3}\pi r^2 h$ respectively.)

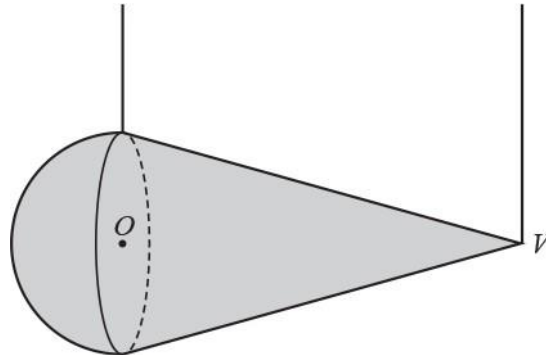
[5]



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S is suspended from two light vertical strings, one attached to V and the other attached to a point on the circumference of the joint circular face, and hangs in equilibrium with OV horizontal (see diagram).

- (c) The weight of S is W . Find the magnitudes of the tensions in the strings in terms of W . [3]

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- 7 The region bounded by the x -axis and the curve $y = ax(2 - x)$, where a is a constant, is occupied by a uniform lamina L_1 (see Fig. 1). Units on the axes are metres.

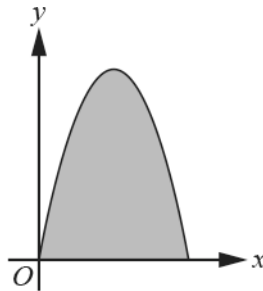


Fig. 1

- (a) Write down the value of the x -coordinate of the centre of mass of L_1 .

[1]



- (b) Show that the y -coordinate of the centre of mass of L is $\frac{2}{5}a$.

[5]

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The mass of L_1 is M kg. A uniform rectangular lamina of width 2 m and height a m is made from a different material from that of L_1 and has a mass of $2M$ kg. A new lamina, L_2 , is formed by joining the straight edge of L_1 to an edge of the rectangular lamina of length 2 m (see Fig. 2).

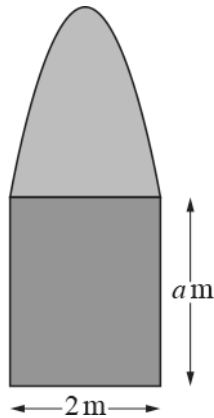
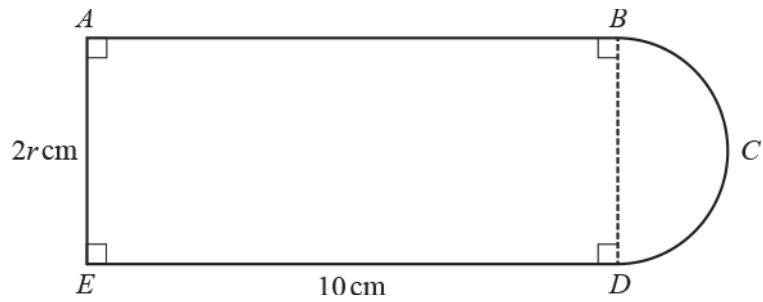


Fig. 2

L_2 is freely suspended from one of its right-angled corners and hangs in equilibrium with its edge of length 2 m making an angle of 20° with the horizontal.

(c) Find the value of a , giving your answer correct to 3 significant figures.

[4]



A uniform lamina $ABCDE$ consists of a rectangle $ABDE$ and a semicircle BCD joined along the common edge BD . $AB = DE = 10$ cm and $AE = BD = 2r$ cm (see diagram).

- (i) Show that the distance of the centre of mass of the lamina from AE is

$$\frac{4r^2 + 30\pi r + 600}{3(40 + \pi r)} \text{ cm.}$$

[5]



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The lamina is freely suspended from B and hangs in equilibrium. The angle between AB and the downward vertical at B is θ , where $\tan \theta = \frac{1}{4}$.

(ii) Show that r satisfies the equation

$$(3\pi + 1)r^2 + 120r - 150 = 0,$$

and hence find r .

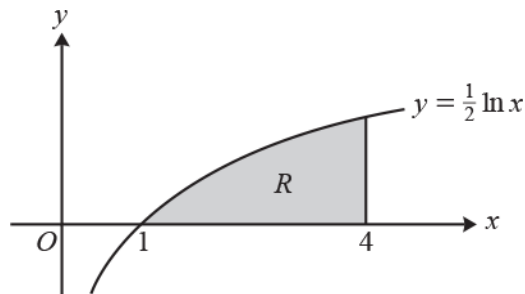
[5]



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The diagram shows the curve with equation $y = \frac{1}{2} \ln x$. The region R , shaded in the diagram, is bounded by the curve, the x -axis and the line $x = 4$. A uniform solid of revolution is formed by rotating R completely about the y -axis to form a solid of volume V .

- (i) Show that $V = \frac{1}{4}\pi(64 \ln 2 - 15)$. [4]

(ii) Find the exact y -coordinate of the centre of mass of the solid.

[4]



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- 10 A uniform rod with centre C has mass $2M$ and length $4a$. The rod is free to rotate in a vertical plane about a smooth fixed horizontal axis passing through a point A on the rod, where $AC = ka$ and $0 < k < 2$. The rod is making small oscillations about the equilibrium position with period T .

- (i) Show that $T = 2\pi \sqrt{\frac{a}{3g} \left(\frac{4+3k^2}{k} \right)}$. (You may assume the standard formula $T = 2\pi \sqrt{\frac{I}{mgh}}$ for the period of small oscillations of a compound pendulum.) [4]



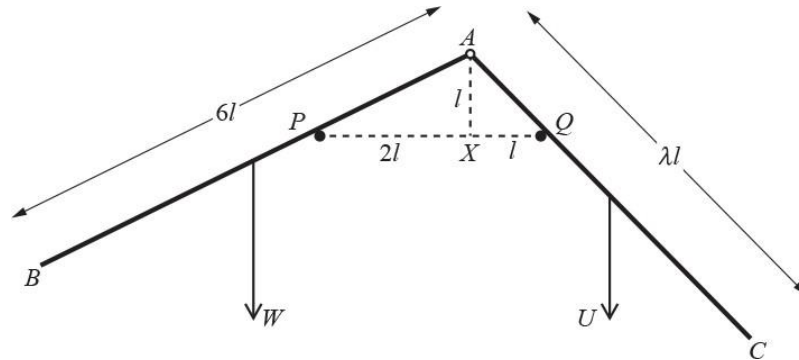
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- (ii) Hence find the value of k^2 for which the period of oscillations is least. [3]

11



Two uniform rods, AB and AC , are freely jointed at A . The weight of AB is W and its length is $6l$; the weight of AC is U and its length is λl , where λ is a constant. The rods rest in equilibrium in a vertical plane on two small smooth pegs, P and Q , which are a distance $3l$ apart at the same horizontal level. The point X is a distance l vertically below A and lies on the line joining P and Q such that $PX = 2l$ and $XQ = l$ (see diagram).

- (i) Show that the magnitude of the force acting on the rod AB at P is $1.2W$, and express the magnitude of the force acting on the rod AC at Q in terms of λ and U . [4]

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- (ii) Find the value of the constant k for which $U = kW$. Find also the value of λ . [7]

- 12 The region bounded by the x -axis, the y -axis, the line $x = \ln 32$ and the curve $y = e^{0.8x}$ for $0 \leq x \leq \ln 32$, is occupied by a uniform lamina (see Fig. 3).

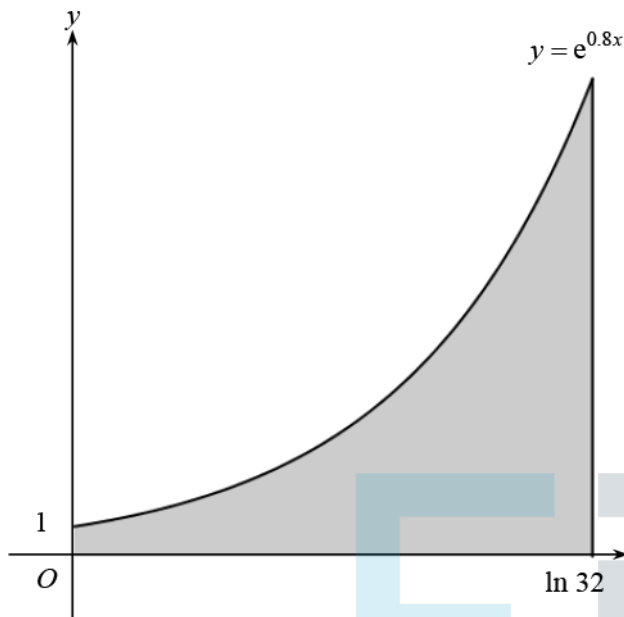


Fig. 3

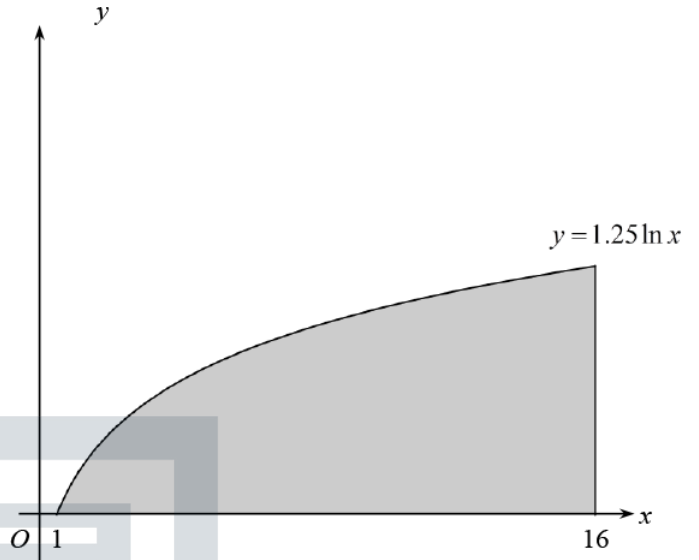


Fig. 4

(a)

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Show that the x-coordinate of the centre of mass of the lamina is given by $\frac{16}{3} \ln 2 - \frac{5}{4}$.

[7]



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(b) Calculate the y -coordinate of the centre of mass of the lamina.

[3]

(c) The region bounded by the x -axis, the line $x = 16$ and the curve $y = 1.25 \ln x$ for $1 \leq x \leq 16$, is occupied by a second uniform lamina (see Fig. 4). By using your answer to part (a) find, to 3 significant figures, the x -coordinate of the centre of mass of this second lamina.

[4]

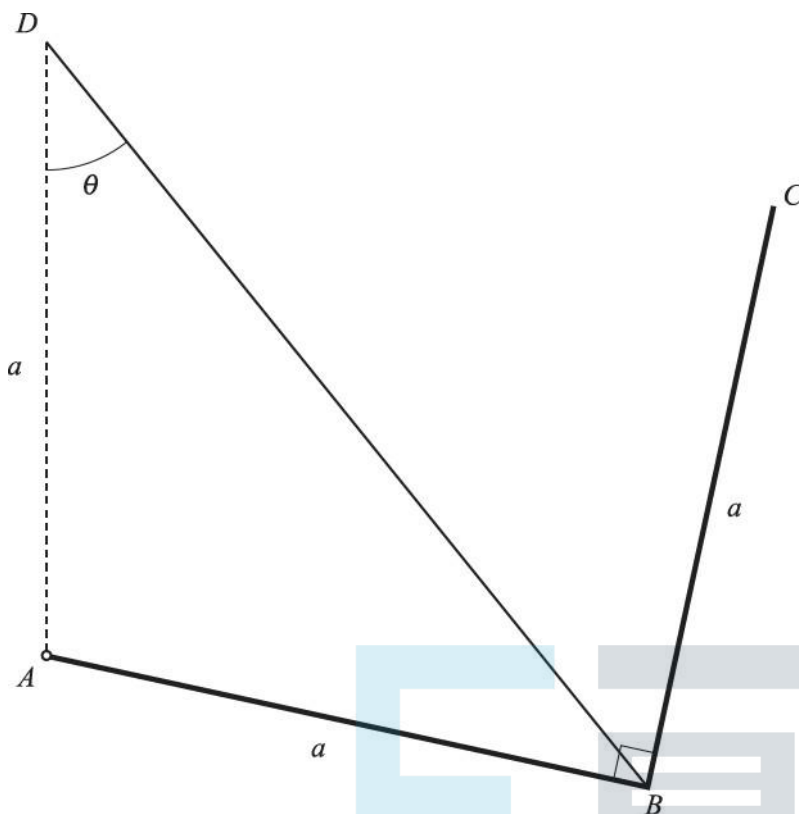


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13



Two uniform rods AB and BC , each of length a and mass m , are rigidly joined together so that AB is perpendicular to BC . The rod AB is freely hinged to a fixed point at A . The rods can rotate in a vertical plane about a smooth fixed horizontal axis through A . One end of a light elastic string of natural length a and modulus of elasticity λmg is attached to B . The other end of the string is attached to a fixed point D vertically above A , where $AD = a$. The string BD makes an angle θ radians with the downward vertical (see diagram).

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(i) Taking D as the reference level for gravitational potential energy, show that the total potential energy V of the system is given by

$$V = \frac{1}{2}mga(\sin 2\theta - 3 \cos 2\theta) + \frac{1}{2}\lambda mga(2 \cos \theta - 1)^2 - 2mga.$$

[5]

(ii) Given that $\theta = \frac{1}{4}\pi$ is a position of equilibrium, find the exact value of λ .

[4]

- (iii) Find $\frac{d^2V}{d\theta^2}$ and hence determine whether the position of equilibrium at $\theta = \frac{1}{4}\pi$ is stable or unstable.

[4]



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- 14 The region bounded by the curve $y = 2e^{\frac{1}{2}x}$ for $0 \leq x \leq 2$, the x -axis, the y -axis and the line $x = 2$, is occupied by a uniform lamina.
- (i) Find the exact value of the y -coordinate of the centre of mass of the lamina.



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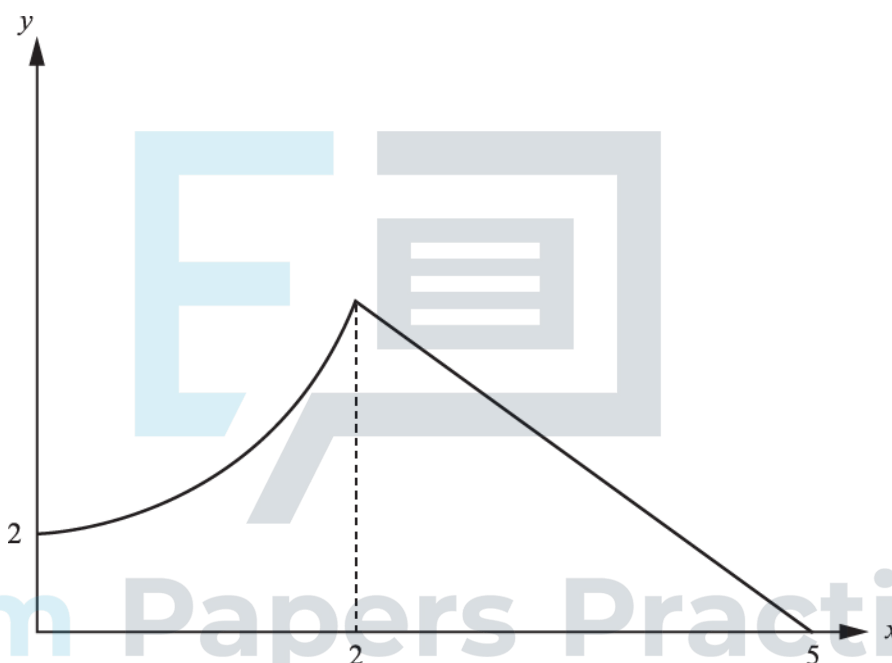
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[6]

As shown in the diagram below, a uniform lamina occupies the closed region bounded by the x -axis, the y -axis and the curve $y = f(x)$ where

$$f(x) = \begin{cases} 2e^{\frac{1}{2}x} & 0 \leq x \leq 2, \\ \frac{2}{3}(5-x)e & 2 \leq x \leq 5. \end{cases}$$



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(ii) Find the exact value of the x -coordinate of the centre of mass of the lamina.

[7]

15 The region bounded by the x -axis, the lines $x = 1$ and $x = 2$, and the curve $y = kx^2$, where k is a positive constant, is occupied by a uniform lamina.

(i) Find the exact x -coordinate of the centre of mass of the lamina.



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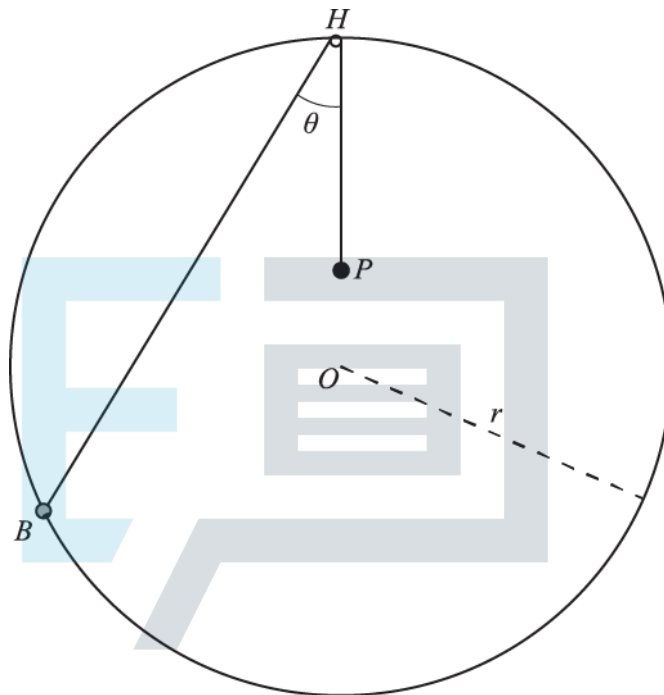
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[6]

(ii) Given that the x - and y -coordinates of the centre of mass of the lamina are equal, find the exact value of k .

[4]

16



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A smooth circular wire hoop, with centre O and radius r , is fixed in a vertical plane. The highest point on the wire is H . A small bead B of mass m is free to move along the wire. A light inextensible string of length a , where $a > 2r$, has one end attached to the bead. The other end of the string passes over a small smooth pulley at H and carries at its end a particle P of mass λm , where λ is a positive constant. The part of the string HP is vertical and the part of the string BH makes an angle θ radians with the downward vertical where $0 \leq \theta \leq \frac{1}{3}\pi$ (see diagram). You may assume that P remains above the lowest point of the wire.

- (i) Taking H as the reference level for gravitational potential energy, show that the total potential energy V of the system is given by

$$V = mg(2\lambda r \cos\theta - 2r \cos^2\theta - \lambda a).$$

[5]

- (i) Find the set of possible values of λ so that there is more than one position of equilibrium.



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[4]

(ii) For the case $\lambda = \frac{3}{2}$, determine whether each equilibrium position is stable or unstable.

- 17 The region bounded by the y -axis and the curves $y = \sin 2x$ and $y = \sqrt{2} \cos x$ for $0 \leq x \leq \frac{1}{4}\pi$ is occupied by a uniform lamina. Find the exact value of the x -coordinate of the centre of mass of the lamina.



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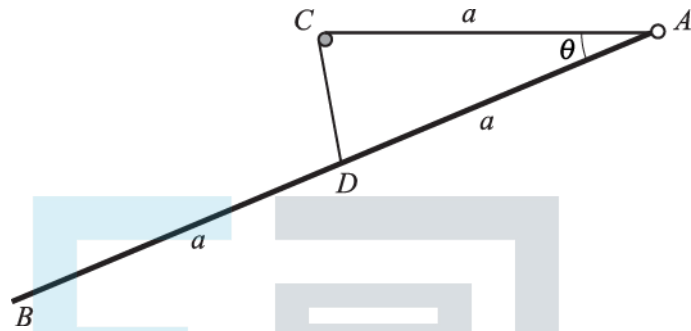
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[6]

[8]

18



A uniform rod AB has mass m and length $2a$. The rod can rotate in a vertical plane about a smooth fixed horizontal axis passing through A . One end of a light elastic string of natural length a and modulus of elasticity $\sqrt{3}mg$ is attached to A . The string passes over a small smooth fixed pulley C , where AC is horizontal and $AC = a$. The other end of the string is attached to the rod at its mid-point D . The rod makes an angle θ below the horizontal (see diagram).

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- (i) Taking A as the reference level for gravitational potential energy, show that the total potential energy V of the system is given by

$$V = mga(\sqrt{3} - \sin \theta - \sqrt{3} \cos \theta).$$

[4]

- (ii) Show that $\theta = \frac{1}{6}\pi$ is a position of stable equilibrium for the system.

[5]



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The system is making small oscillations about the equilibrium position.

(iii) By differentiating the energy equation with respect to time, show that

$$\frac{4}{3}a\ddot{\theta} = g(\cos\theta - \sqrt{3}\sin\theta).$$

[4]

(iv) Using the substitution $\theta = \phi + \frac{1}{6}\pi$, show that the motion is approximately simple harmonic, and find the approximate period of the oscillations.

[6]

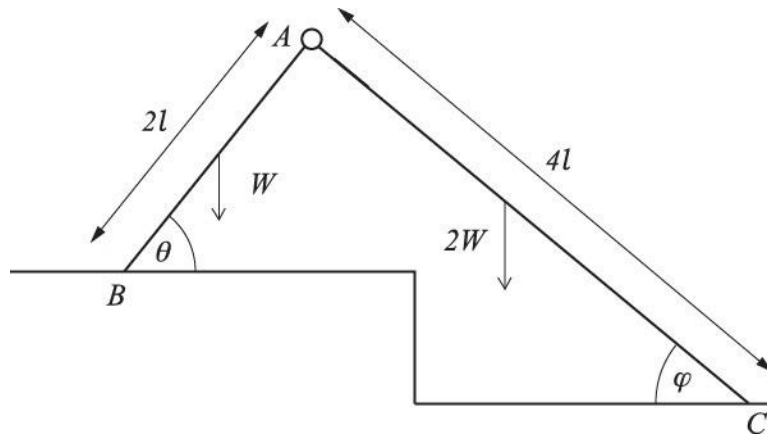
- 19 A straight rod AB has length a . The rod has variable density, and at a distance x from A its mass per unit length is given by $k\left(4 - \sqrt{\frac{x}{a}}\right)$, where k is a constant. Find the distance from A of the centre of mass of the rod.

[7]

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Two uniform rods AB and AC are freely jointed at A . Rod AB is of length $2l$ and weight W ; rod AC is of length $4l$ and weight $2W$. The rods rest in equilibrium in a vertical plane on two rough horizontal steps, so that AB makes an angle of θ with the horizontal, where $\sin \theta = \frac{4}{5}$ and AC makes an angle of ϕ with the horizontal, where $\sin \phi = \frac{3}{5}$ (see diagram). The force of the step acting on AB at B has vertical component R and horizontal component F .

- (i) By taking moments about A for the rod AB , find an equation relating W , R and F .

[3]

- (ii) Show that $R = \frac{73}{50}W$, and find the vertical component of the force acting on AC at C .

[6]

- (iii) The coefficient of friction at B is equal to that at C . Given that one of the rods is on the point of slipping, explain which rod this must be, and find the coefficient of friction.

[4]

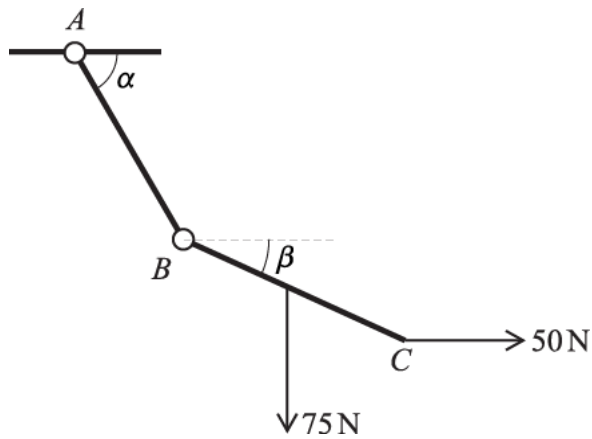


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21



Two uniform rods AB and BC , each of length $2L$, are freely jointed at B , and AB is freely jointed to a fixed point at A . The rods are held in equilibrium in a vertical plane by a light horizontal string attached at C . The rods AB and BC make angles α and β to the horizontal respectively. The weight of rod BC is 75 N , and the tension in the string is 50 N (see diagram).

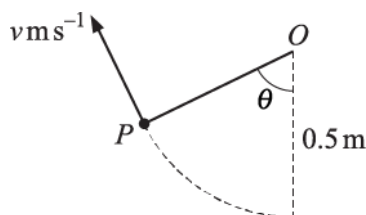
(i) Show that $\tan \beta = \frac{3}{4}$. [3]

(ii) Given that $\tan \alpha = \frac{12}{5}$, find the weight of AB . [5]

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One end of a light inextensible string of length 0.5 m is attached to a fixed point O . A particle P of mass 0.2 kg is attached to the other end of the string. P is projected horizontally from the point 0.5 m below O with speed $u \text{ ms}^{-1}$. When the string makes an angle of θ with the downward vertical the particle has speed $v \text{ ms}^{-1}$ (see diagram).

- (i) Show that, while the string is taut, the tension, $T \text{ N}$, in the string is given by

$$T = 5.88 \cos \theta + 0.4u^2 - 3.92.$$

[5]

- (ii) Find the least value of u for which the particle will move in a complete circle.

- (iii) If in fact $u = 3.5 \text{ ms}^{-1}$, find the speed of the particle at the point where the string first becomes slack.

[3]

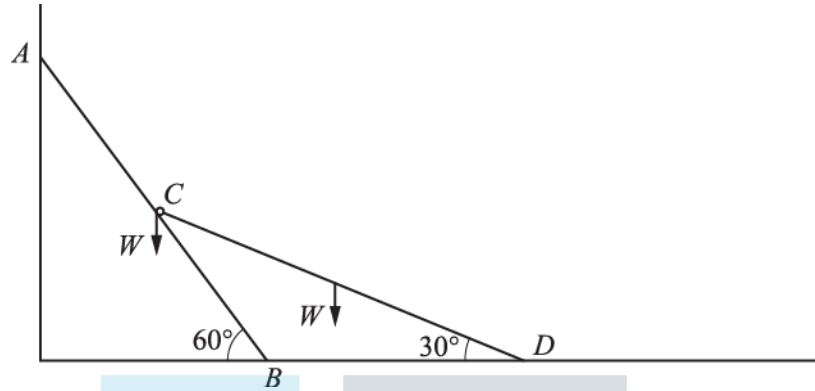
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[4]

- 23 A uniform rod AB , of weight W and length $2l$ is in equilibrium at 60° to the horizontal with A resting against a smooth vertical plane and B resting on a rough section of a horizontal plane. Another uniform rod CD , of length $\sqrt{3}l$ and weight W , is freely jointed to the mid-point of AB at C ; its other end D rests on a smooth section of the horizontal plane. CD is inclined at 30° to the horizontal (see diagram).



- (i) Show that the force exerted by the horizontal plane on CD is $\frac{1}{2}W$. Find the normal component of the force exerted by the horizontal plane on AB .

[5]

(ii) Find the magnitude and direction of the force exerted by CD on AB .



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[3]

(iii) Given that AB is in limiting equilibrium, find the coefficient of friction between AB and the horizontal plane.

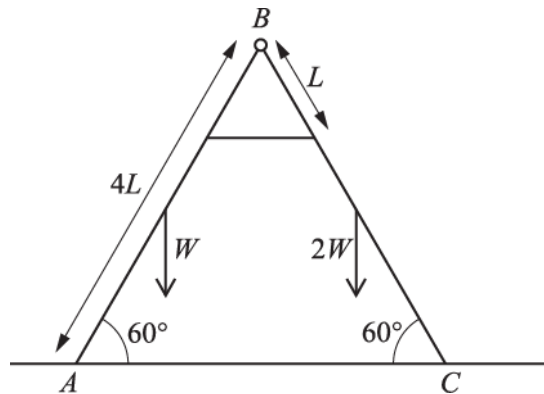
[5]



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Two uniform rods AB and BC , each of length $4L$, are freely jointed at B , and rest in a vertical plane with A and C on a smooth horizontal surface. The weight of AB is W and the weight of BC is $2W$. The rods are joined by a horizontal light inextensible string fixed to each rod at a point distance L from B , so that each rod is inclined at an angle of 60° to the horizontal (see diagram).

- (i) By considering the equilibrium of the whole body, show that the force acting on BC at C is $1.75W$ and find the force acting on AB at A .

[4]

- (ii) Find the tension in the string in terms of W .

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[4]

- (iii) Find the horizontal and vertical components of the force acting on AB at B , and state the direction of the component in each case.

[3]

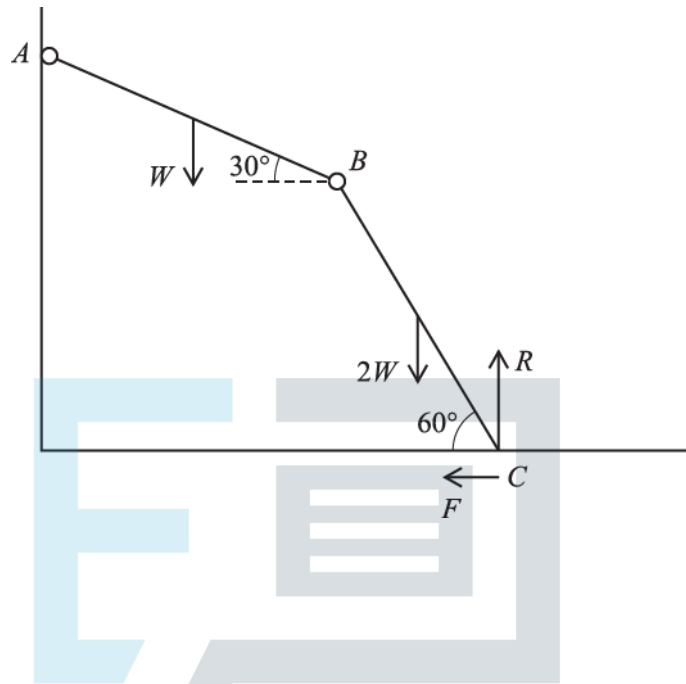


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- 25 Two uniform rods AB and BC , each of length $2l$, are freely jointed at B . The weight of AB is W and the weight of BC is $2W$. The rods are in a vertical plane with A freely pivoted at a fixed point and C resting in equilibrium on a rough horizontal plane. The normal and frictional components of the force acting on BC at C are R and F respectively. The rod AB makes an angle 30° to the horizontal and the rod BC makes an angle 60° to the horizontal (see diagram).



- (i) By considering the equilibrium of rod BC , show that $W + \sqrt{3}F = R$.

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[2]

- (ii) By taking moments about A for the equilibrium of the whole system, find another equation involving W , F and R .

[4]

- (iii) Given that the friction at C is limiting, calculate the value of the coefficient of friction at C .

[5]



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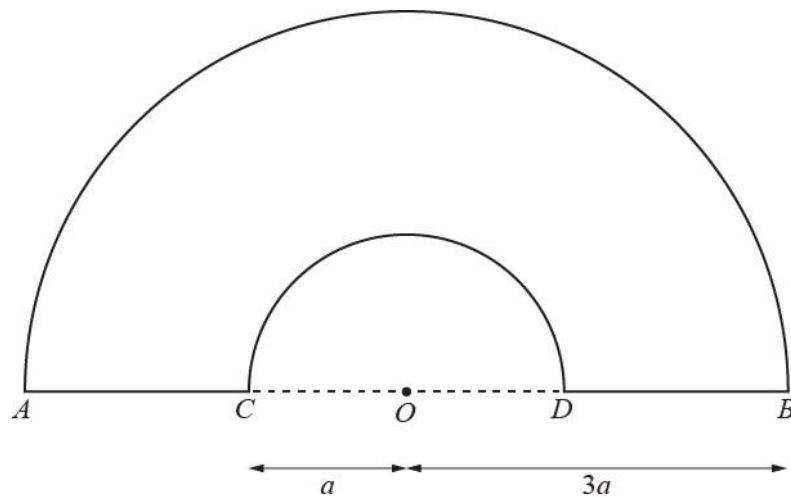


Fig. 1

A uniform lamina $ABDC$ is bounded by two semicircular arcs AB and CD , each with centre O and of radii $3a$ and a respectively, and two straight edges, AC and DB , which lie on the line AOB (see Fig. 1).

- (i) Show that the distance of the centre of mass of the lamina from O is

[5]

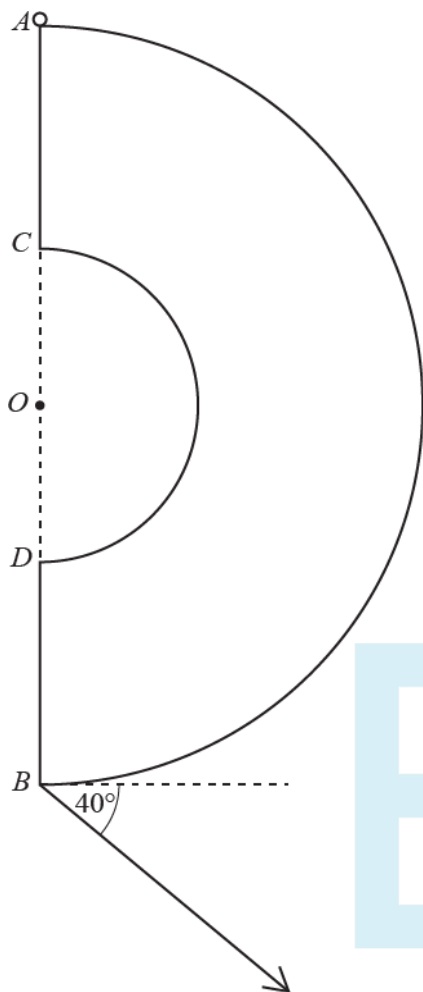


Fig. 2

The lamina has mass 3 kg and is freely pivoted to a fixed point at A . The lamina is held in equilibrium with AB vertical by means of a light string attached to B . The string lies in the same plane as the lamina and is at an angle of 40° below the horizontal (see Fig. 2).

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(ii) Calculate the tension in the string.

[3]

(iii) Find the direction of the force acting on the lamina at A.

[4]



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- 27 A uniform ladder AB , of weight W and length $2a$, rests with the end A in contact with rough horizontal ground and the end B resting against a smooth vertical wall. The ladder is inclined at an angle θ to the horizontal, where $\sin \theta = \frac{3}{5}$. A man of weight $6W$ is standing on the ladder at a distance x from A and the system is in equilibrium.

- (i) Show that the magnitude of the frictional force exerted by the ground on the ladder is $\frac{3}{2}W$. [5]

The coefficient of friction between the ladder and the ground is $\frac{1}{3}$.

- (ii) Find, in terms of a , the greatest value of x for which the system is in equilibrium.



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[3]

The bottom of the ladder A is moved closer to the wall so that the ladder is now inclined at an angle α to the horizontal. The man of weight $6W$ can now stand at the top of the ladder B without the ladder slipping.

(iii) Find the least possible value of $\tan \alpha$.

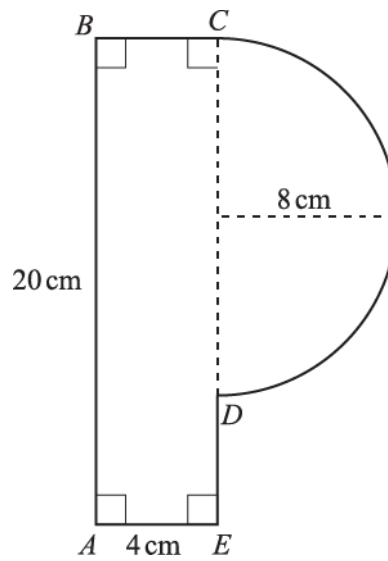
[3]



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A uniform solid prism has cross-section $ABCDE$ in the shape of a rectangle measuring 20 cm by 4 cm joined to a semicircle of radius 8 cm as shown in the diagram. The centre of mass of the solid lies in this cross-section.

- (i) Find the distance of the centre of mass of the solid from AB .

[5]

The solid is placed with AE on rough horizontal ground (so the object does not slide) and is in equilibrium with a horizontal force of magnitude 4 N applied along CB .

(ii) Find the greatest and least possible values for the weight of the solid.

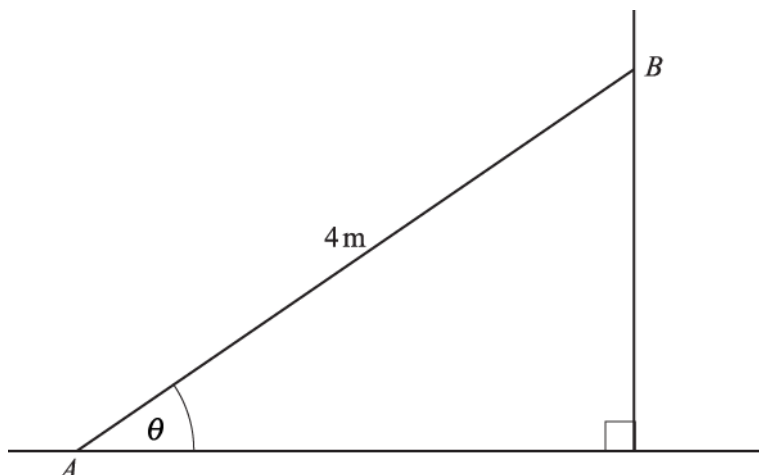
[5]



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A uniform ladder AB of weight WN and length 4 m rests with its end A on rough horizontal ground and its end B against a smooth vertical wall. The ladder is inclined at an angle θ to the horizontal where $\tan \theta = \frac{1}{2}$ (see diagram). A small object S of weight $2WN$ is placed on the ladder at a point C , which is 1 m from A . The coefficient of friction between the ladder and the ground is μ and the system is in limiting equilibrium.

- (i) Show that $\mu = \frac{2}{3}$

[6]

A small object of weight aW N is placed on the ladder at its mid-point and the object S of weight $2WN$ is placed on the ladder at its lowest points A.

(ii) Given that the system is in equilibrium, find the set of possible values of a .

[5]

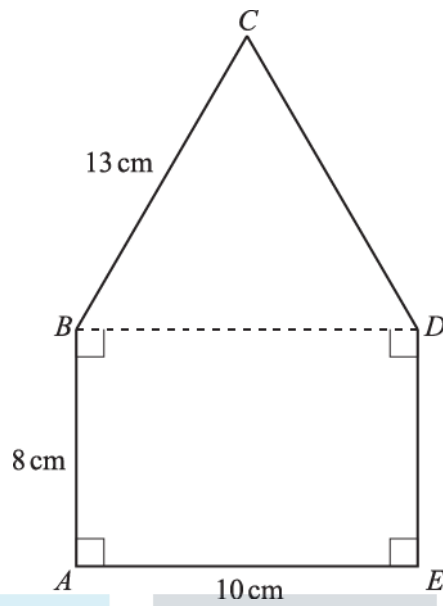


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30



A uniform lamina $ABCDE$ consists of a rectangle $ABDE$ and an isosceles triangle BCD joined along their common edge. $AB = DE = 8\text{ cm}$, $AE = BD = 10\text{ cm}$ and $BC = CD = 13\text{ cm}$ (see diagram).

- (i) Find the distance of the centre of mass of the lamina from AE .

[5]

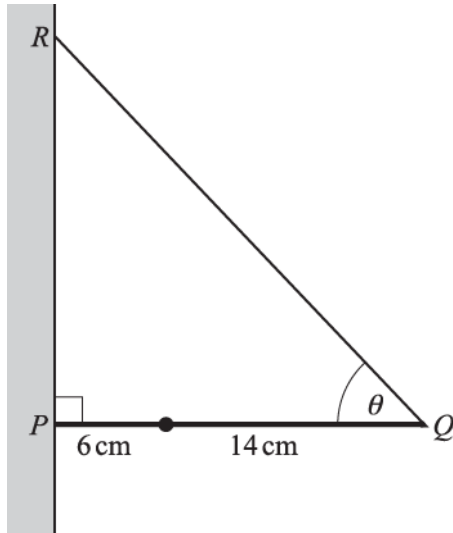
- (ii) The lamina is freely suspended from B and hangs in equilibrium. Calculate the angle that BD makes with the vertical.

[3]

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31



A uniform rod PQ has weight 18 N and length 20 cm. The end P rests against a rough vertical wall. A particle of weight 3 N is attached to the rod at a point 6 cm from P . The rod is held in a horizontal position, perpendicular to the wall, by a light inextensible string attached to the rod at Q and to a point R on the wall vertically above P , as shown in the diagram. The string is inclined at an angle θ to the horizontal, where $\sin \theta = \frac{3}{5}$. The system is in limiting equilibrium.

- (i) Find the tension in the string.

[3]

(ii) Find the magnitude of the force exerted by the wall on the rod.



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[4]

(iii) Find the coefficient of friction between the wall and the rod.

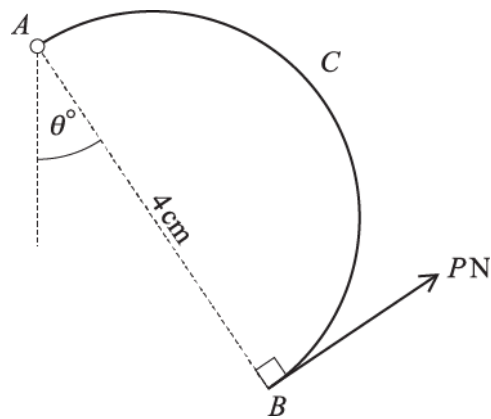
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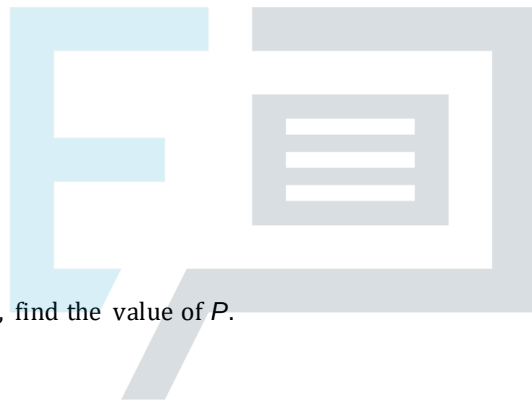
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A uniform semicircular arc ACB is freely pivoted at A . The arc has mass 0.3 kg and is held in equilibrium by a force of magnitude $P\text{ N}$ applied at B . The line of action of this force lies in the same plane as the arc, and is perpendicular to AB . The diameter AB has length 4 cm and makes an angle of θ° with the downward vertical (see diagram).

- (i) Given that $\theta = 0$, find the magnitude of the force acting on the arc at A .



[6]

(ii) Given instead that $\theta = 30$, find the value of P .

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[4]

- 33 A solid uniform cone has height 8 cm, base radius 5 cm and mass 4 kg. A uniform conical shell has height 10 cm, base radius 5 cm and mass 0.4 kg. The two shapes are joined together so that the circumferences of their circular bases coincide.

(i) Find the distance of the centre of mass of the shape from the common circular base.

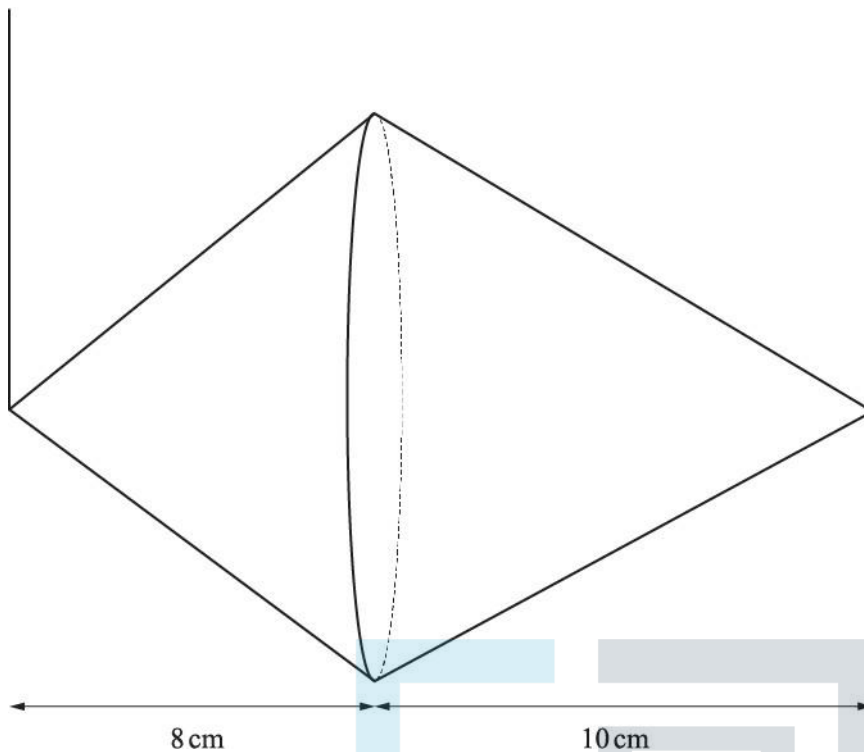
[4]



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The object is suspended with a string attached to the vertex of the cone and another string attached to the vertex of the conical shell. The object is in equilibrium with the strings vertical and the axis of symmetry of the object horizontal (see diagram).

(ii) Find the tension in each string.

[4]

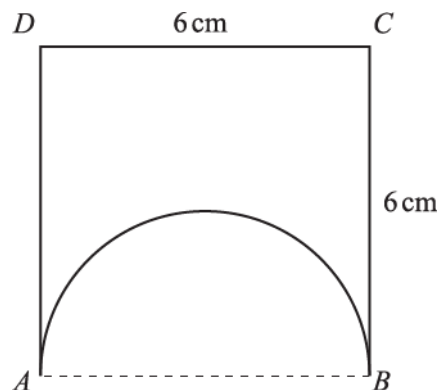
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34



A uniform square lamina $ABCD$ of side 6 cm has a semicircular piece, with AB as diameter, removed (see diagram).

- (i) Find the distance of the centre of mass of the remaining shape from CD .

[6]

The remaining shape is suspended from a fixed point by a string attached at C and hangs in equilibrium.

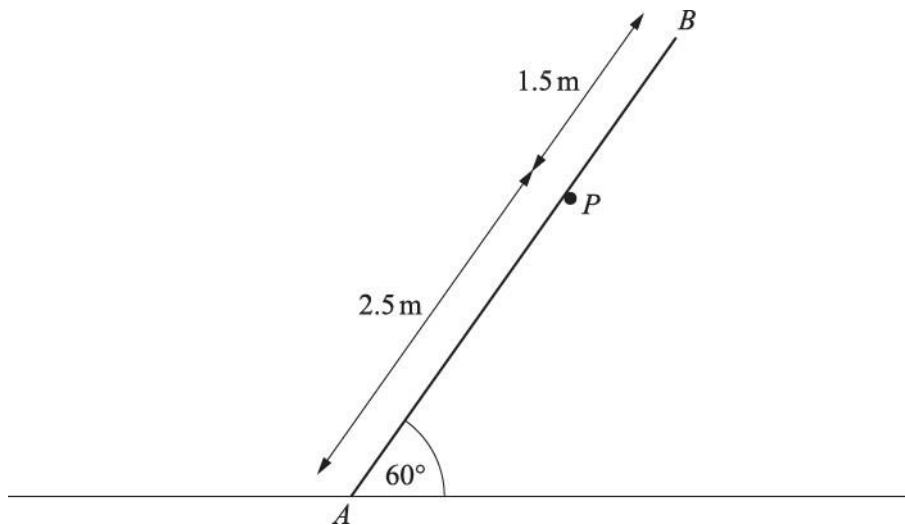
- (ii) Find the angle between CD and the vertical.

[2]

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A uniform rod AB , of mass 3 kg and length 4 m , is in limiting equilibrium with A on rough horizontal ground. The rod is at an angle of 60° to the horizontal and is supported by a small smooth peg P , such that the distance AP is 2.5 m (see diagram). Find

- (i) the force acting on the rod at P ,

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[3]

(ii) the coefficient of friction between the ground and the rod.

[5]



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